



A framework for reactive transport modeling using FEniCS–Reaktoro: governing equations and benchmarking results

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Abstract

Reactive transport codes are widely applied in geoscience to predict or reconstruct spatial and temporal evolution of geochemical systems. To provide an accurate description of natural systems at different spatial and temporal scales, the reactive transport code has to deal with coupling of different physical and chemical phenomena. Many reactive transport codes have been developed in the past and each of these codes has specific strengths and limitations. Here, we present a new versatile reactive transport framework based on the FEniCS equations solver and the chemical solver Reaktoro. This development was motivated by the need for an advanced open-source tool allowing user-friendly modeling environment and, at the same time, full control over the numerical methods. Unlike most of the currently available codes, the developed FEniCS–Reaktoro framework offers full flexibility in setting up the reactive transport simulations of arbitrary complexity in terms of process couplings, simulation domain geometry and the boundary conditions applied. The simulations are setup using a simple high-level scripting language intuitively linked to the equation based model definition without the need of advanced programming skills. The chemical solver Reaktoro allows thermodynamic modeling of multicomponent multiphase system with several fluids and solid phases, including highly non-ideal solid solutions. The coupling of transport and chemistry is implemented using the sequential non-iterative approach (SNIA) in which the transport of the aqueous components and the chemical reactions are solved in two consequent steps. The flexibility and results of the FEniCS–Reaktoro framework are demonstrated against several widely accepted reactive transport benchmarks.

Keywords Reactive transport · Electrochemical transport · Multicomponent diffusion · Finite element method · Porous media · Gibbs energy minimization · Operator splitting approach

1 Introduction

Reactive transport modeling is widely applied to study the temporal and spatial evolution of geochemical conditions in natural system in various geoscientific fields [1]. To provide an accurate description of natural systems at different spatial and temporal scales, the reactive transport code has to deal with the coupling of different physical and chemical phenomena, such as heat transfer [2], advective fluid transport in porous and fractured media [3], diffusive and electrochemical migration of chemical species [4], electromigration [5, 6], mechanical stress [7], and heterogeneous chemical reactions [8, 9].

To allow for full control in the system description, a general purpose code should thus be:

- Able to solve chemical systems of realistic complexity
- Able to model physical processes for arbitrary complex geometries
- Numerically accurate and robust
- Easily scalable in terms of process coupling and system complexity
- Compatible with HPC architecture
- Well documented and preferably available as open-source

Table 1 compares the capabilities of popular reactive transport codes currently available to the scientific community. Each code listed in Table 1 has specific advantages and limitations related to the specific implementation of the modeling approach. These are analyzed in detail in the following sections.

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Table 1 Reactive transport codes comparison

Code	Fully open-source	Numerical method ¹	Dimensions	Mass transport PDE ²	Electrochemical migration	Chemical method ³	Compatible with HPC	Numerical approach ⁴
CrunchFlow [10]	Yes	FV	1,2,3D	BI	Yes	LMA	No	GI, OS
Comsol-Phreeqc iCP [11]	No	FE	1,2,3D	UD	No	LMA	Yes	OS
Comsol-Phreeqc [12]	No	FE	1,2,3D	UD	Yes	LMA	No	OS
Comsol-GEMS [13]	No	FE	1,2,3D	UD	Yes	GEM	Yes	OS
CORE ^{2D} [14]	No	FE	1,2D	BI	No	LMA	No	OS
CSMP++GEM [15]	No	FE, FV	1,2,3D	BI	No	GEM	Yes	OS
DuCOM-Phreeqc [16]	No	FE	ID	BI	Yes	LMA	No	OS
Frachem [2]	No	FE, FV	1,2D	BI	No	LMA	No	OS
GeoSysBRNS [9]	No	FE	1,2,3D	BI	No	LMA	Yes	OS
HP1/HPx [17]	No	FE	1,2,3D	BI	No	LMA	Yes	OS
Matlab-IPhreeqc [6]	No	FE, FV	1,2D	BI	Yes	LMA	No	OS
MIN3P [18]	No	FV	1,2,3D	BI	Yes	LMA	No	GI
The Geochemist's Workbench [19]	No	FD	1,2D	BI	No	LMA	Yes	OS
OGS-Chemapp [20]	No	FE	1,2,3D	BI	No	GEM	No	OS
OGS-GEM [21]	Yes	FE	1,2,3D	BI	No	GEM	Yes	OS
OGS-IPhreeqc [22]	Yes	FE	1,2,3D	BI	No	LMA	Yes	OS
Orchestra [23]	No	MC	1,2D	UD	Yes	LMA	Yes	OS
Pflotran [24]	Yes	FV	1,2,3D	BI	No	LMA	Yes	GI, OS
Phreeqc [25]	Yes	MC	ID	BI	Yes	LMA	No	OS
Proost [26]	No	FE	1,2,3D	BI	No	LMA	No	OS
ReactMiCP [27]	Yes	FE	ID	BI	No	LMA	Yes	OS
ReacTran [28]	Yes	FD	1,2,3D	UD	No	RR	Yes	GI
Toughreact [29]	No	FV	1,2,3D	BI	No	LMA	Yes	OS
TReacLab ⁵ [30]	No	FD	1,2D	UD	Yes	LMA	No	OS

¹FE denotes finite elements, FV denotes finite volumes, FD denotes finite differences, and MC denotes mixing cell

²BI denotes built-in and UD denotes user-defined

³LMA denotes law of mass action, GEM denotes Gibbs energy minimization and RR denotes reaction rates

⁴GI denotes global implicit and OS denotes operator splitting

⁵Since the purpose of TReacLab is to provide a generic operator splitting interface, its definitions vary according to the transport and chemical solvers used

The selection of the simulation tool should be based on a careful evaluation of code functionality to assure that the simulation tools meet the user's requirements. In addition to a broad range of physical phenomena, a number of recently developed codes also support parallelism and provide HPC compatibility [31–33]. The numerical performance and scalability of the codes are essential criteria when realistic 2D and 3D natural systems are considered.

Chemical solvers based on law of mass action (LMA) (e.g., equilibrium constant of chemical reactions) are most commonly used to calculate equilibrium speciation in reactive transport codes due to the simplicity of the algorithm [34]. The LMA solvers have limitations when multicomponent phases (e.g., solid solutions, non-ideal liquid, and gaseous phases) need to be considered. The Gibbs energy minimization (GEM) algorithm [8] is the method of choice for the simulations of multiphase systems and is based on the standard chemical potentials. Both LMA and GEMS have pros and cons when practical applications are considered. Both approaches are based on different algorithms and assumptions but are equivalent from the theoretical point of view and must, therefore, provide the same result. Recently, Leal et al. [35] developed an innovative approach referred to as xLMA method which combines the basic algorithms used in the LMA and GEM approach resulting in substantial improvement of the stability and performance of the chemical solver.

Depending on the code design, the mass transport equations can be built-in into the code or defined by the user via an external interface or by a high-level scripting language. Most commonly, the partial differential equations (PDE) are hard-coded within the core routines and not accessible or changeable by the user, as in [25]. In contrast, in a more flexible approach, some codes provide a scripting or graphical interface that allows the user to define the mass transport equation, as in [23, 36].

Coupling of reactions and transport can be implemented using the operator splitting (OS) approach [37, 38], or the global implicit (GI) method [39, 40]. In practice, both methods become intractable for the complex system due to the complexity of numerical implementation and increasing computer memory requirements. The pros and cons of each approach have been discussed extensively in the literature, as in [41–43].

The characteristic time scale of physical and chemical phenomena or spatial heterogeneities in natural systems can span over several orders of magnitude [44]. Numerical simulations of such systems are very challenging because the appropriate integration time and the spatial discretization of the domain are typically limited by the fastest small-scale phenomena. Numerical codes typically hit the performance limits when pore-scale phenomena [45] controls the processes at the field scale [46, 47]. Particularly challenging

are the simulations of systems with strong chemical gradients such as nuclear waste disposal sites or geochemical installations [48, 49].

Transport equations can be solved by different numerical methods: finite elements (FE) [50–52], finite differences (FD), or finite volumes (FV) [53]. FE is well suited for dealing with complex geometries and provides extra stability by using small elements where the function is changing rapidly and bigger elements elsewhere. The FE method has disadvantages as far as the mass and the energy conservation are concerned. The FV method ensures local mass and energy conservation but requires more efforts to describe irregular geometries. For the sake of completeness, it is worth mentioning an alternative approach based on lattice Boltzmann method [54]. The particular strength of the LB method is the exceptional stability of the LB equations in a wide range of hydrodynamic regimes and a straightforward implementation of a parallelization algorithm.

The ability of a code to allow the description of materials properties by different constitutive relationships, i.e., porosity [55], tortuosity [56], diffusivity [57], and boundary conditions [58], is critical for successful implementation of system setup. The possibility to allow user-controlled parametrization and consideration of more specific processes and interaction, e.g., anisotropic diffusion [59] or porosity–permeability relation [60], are highly desirable features which are not commonly available in most of the codes.

The numerical performance of the code and the compatibility with high-performance computing (HPC), as in [11, 32, 61], deliver better performance when treating a large system of equations necessary for describing the physical and chemical phenomena. The numerical approaches take advantage of the usage of HPC during the multithreaded node-oriented chemical calculation, in the OS approach, or by an efficient solution of the large equation system, in the GI approach. Since the speed of the clock is stagnated at 2–3 GHz, the further increase of code performance needs to be improved by taking advantage of multicore architecture CPUs and massively parallel HPC facilities. The multi-threading is a simple way of improving the code performance but is limited essentially to the shared memory architecture. Most of the modern HPC facilities rely on distributed memory architecture. Use of such a resources prerequisite decomposition of the simulated domain and explicit data exchange between subdomains.

The usage of commercial closed source (or proprietary) codes, for example, [19, 36], often involves costs for licensing and subscriptions that easily exceed the costs of research projects and make them accessible only to a limited number of researchers. Commercial codes' usability is based on a predefined application programming interface (API) which typically limits their flexibility and transparency. Nonetheless, for the most part, they offer good quality

documentation and support. The open-source codes, like [21, 28], are typically developed at research institutes and universities as part of scientific projects. The development is often discontinued and many times the projects are short on documentation and reduced support. Most of the open-source codes have a potential to benefit from collaborative development and intensive benchmarking by users, as in the SeS Bench (Subsurface Environmental Simulation benchmark) [62]. Exemplary, the potential of the NumFOCUS <https://numfocus.org/>, which has been recognized by research and industry communities in multiple projects [63–66].

2 FEniCS–Reaktoro framework

In this work, we present a new framework for reactive transport simulations called FEniCS–Reaktoro, able to deal with complex multicomponent heterogeneous systems. It follows an object-oriented (OO) approach and uses Python scripting interface for the system setup and coupling of the PDE solver FEniCS [67] with the chemical solver Reaktoro [68]. It allows fast and flexible system setup with a moderate requirement of programming skills. The framework is suited for systems with different spatial and temporal scales such as modeling of a short-term laboratory scale and medium-term field-scale reactive transport experiments, or large-scale reservoir modeling. In the following section, we present a detailed description of the tools used, design, architecture, and capabilities of the FEniCS–Reaktoro framework. We also describe the equations used in the benchmarks as well as the description of the model setup. Numerical benchmarks and applications are presented to demonstrate the versatility of the FEniCS–Reaktoro framework. The combination of such features is not currently available in another open-source reactive transport code, which makes FEniCS–Reaktoro a unique and thriving modeling platform.

2.1 FEniCS

FEniCS is a popular open-source computing framework for solving PDE with several FE-based families. The significant advantage of using FEniCS, compared to other existing software packages, is the flexibility of an automated and efficient solver for the differential equations. The user has essentially to define the equations, and FEniCS takes care of numeric methods necessary to solve the mathematical problem. With FEniCS, the PDE is expressed in the weak form in a math-like human-readable way called Unified Form Language (UFL). For more details on the weak formulation and UFL, we refer to [69]. FEniCS provide the user with full control on the PDE level and allow a quick

change of the FE family, equation definition (as well as addition of extra terms into the equation), equation solver, and use other geometries and dimensions (1D, 2D, or 3D). FEniCS offers different inbuilt boundary conditions such as Dirichlet, Neumann, Robin, several solvers (i.e., minimal residual method, generalized minimal residual method, PETSc library built-in LU solver, and biconjugate gradient stabilized method), and preconditioners (i.e., incomplete LU factorization, and PETSc algebraic multigrid). FEniCS provides portability to HPC [70] without rewriting existing codes based on the PETSc [71]. The documentation and examples are available at <https://www.fenicsproject.com> and it is available in C++ and Python.

2.2 Reaktoro

Reaktoro is an open-source software for modeling chemically reactive systems [68]. It implements efficient numerical methods for solving chemical equilibrium and kinetics for multiphase systems that are particularly suitable for demanding applications such as reactive transport modeling. Reaktoro takes advantage of the GEM method to define the stable phases in equilibrium and is able to import files containing thermodynamic systems from GEM–Selektor [8] and Phreeqc [25].

Reaktoro can model non-ideal aqueous solutions using Davies, Debye–Huckel, Pitzer, and HKF activity models. For gases, Reaktoro implements cubic equations of state such as Peng–Robinson, Redlich–Kwong, Soave–Redlich–Kwong, and some models specific for certain mixtures, such as [72] for H₂O – CO₂ and [73] for H₂O – CO₂ – CH₄. Reaktoro can also model the non-ideal thermodynamic behavior of solid solutions using, for example, Redlich–Kister solid solution model.

Reaktoro also offers support for the CEMDATA, which is a cement specific database [74], the PSI-Nagra database [75], the SUPCRT92 [76], and the Phreeqc [25] databases. Reaktoro allows the flexibility to model high temperatures and pressures (up to 1000 °C and 5000 bar).

Reaktoro is implemented in C++ and offers a Python interface. Its source code, documentation, and examples are available at <https://www.reaktoro.com>.

3 Mathematical setup

The reference model discussed in the following section describes the single-phase transport of aqueous species in fully saturated porous media with a realistic description of the chemical system. The model includes the following:

- Advection, diffusion, and electrochemical migration of aqueous species in an incompressible flow (aqueous

phase). Including the transport across pores confined by charged mineral surfaces [77, 78] by the Poisson–Nernst–Planck equation

- The multiphase chemical systems, including ternary or higher order non-ideal solid solutions, in which precipitation and dissolution of minerals can alter material properties [79]

The description of the reactive mass transport in a general form, based in [80], is as follows:

$$\frac{\partial c_i}{\partial t} = -\nabla(Vc_i - D_i(\nabla c_i + \frac{z_i F c_i}{R T} \nabla \Phi)) + Q_i \quad (1)$$

and

$$\frac{\partial c_i}{\partial t} = \Gamma(c_1 \dots c_m) \quad (2)$$

where c_i is the volumetric concentration (mol/m³) of the i th transported species, t is time (s), V is the pore velocity (m/s), D_i is the diffusion coefficient (m²/s), z_i is the charge (C), F is the Faraday constant (C/mol), Φ is the electrical potential (V), R is the gas constant (J/mol/K), T is the temperature (K), Q_i is the source/sink term, $\Gamma(c_1 \dots c_m)$ is the chemical change rate term (mol/s), and m is the total number of species in the multi-species system. In the right side of Eq. 1, the first term refers to the advective transport, the second term to the aqueous diffusion, the third to the electrochemical migration, and the last to the source/sink term. The Poisson equation that describes Φ , as in [81], is as follows:

$$\nabla^2 \Phi + \frac{F}{\epsilon} \left(\sum_{i=1}^m z_i c_i + w \right) = 0 \quad (3)$$

where ϵ is the dielectric constant (or relative permittivity) of the material (C/V/m) and w is the charge density of the medium (C/m). The combination of Eqs. 1 and 3 is known as the Poisson–Nernst–Planck (PNP) equations. Such equation system also needs a definition of initial and boundary values for concentrations and electric potential, which are defined according each model's definition. For solving transient transport with FE methods (as implemented in FEniCS), we discretize the above equation system in time and transform it into the so-called weak form. We choose the OS approach with an implicit time discretization scheme due to its stability when coupling transport and chemical solvers using OS [82] in combination with small time step length. The implicit time step scheme, as in [10], is as follows:

$$\frac{\partial c_i}{\partial t} \approx \frac{c^n - c^{n-1}}{\Delta t} \quad (4)$$

where c^n is the current unknown concentration, c^{n-1} is the known concentration from previous time step, and Δt is the time step size. In the OS approach, the transport of the aqueous components and the chemical reactions are solved

in two sequential steps. It allows maximal flexibility for coupling the transport and chemical solvers but requires a time step size that respects the Courant–Friedrichs–Lewy (CFL) condition [83], the von Neumann criteria [84], and the rate-limiting step (RLS) [85] for kinetically controlled reactions. The time step size needs to account correctly the transport rates as well as the equilibrium (fast) and kinetic (slow) reaction rates, and properly address the different orders of magnitude between them (from microseconds up to several days) [86]. Thus, it could be expected thousands or millions of time step iterations from the initial condition until some time of interest according to each model.

The weak formulation is achieved by multiplying the regular PDE form with a test function (v_i) and integrating over the domain. Such test function needs to be smooth, continuous, and differentiable inside the domain. Therefore, the solution of the system becomes numerically easier, if compared to solving the system based on its regular PDE form, also known as strong form.

In this work, all the weak formulations presented are defined for the Lagrange family of FE basis with linear interpolation, and we assume either a constant pore velocity or zero, for cases without advective transport. Alternatively, one could use the discontinuous Galerkin (DG) family of elements [87, 88], which can be done in FEniCS by describing the weak formulation in the appropriate DG form, as in [89, 90].

3.1 Transport solver

Table 2 summarizes the equations implemented and presented in this work:

3.1.1 Poisson–Nernst–Planck equation

In a strongly diffusion dominated transport materials, e.g., low permeability bentonite or Opalinus clay [91], it is possible to neglect the advective term. Assuming the source/sink term zero and combining (1) and 4, we achieve the following:

$$\frac{c^n - c^{n-1}}{\Delta t} = -\nabla \left(-D_i(\nabla c_i + \frac{z_i F c_i}{R T} \nabla \Phi) \right) \quad (5)$$

Table 2 Description of the equations implemented for the benchmarks presented in this work

Transport equation description	Eq.
Poisson–Nernst–Planck (PNP)	Eqs. 7 and 6
Nernst–Planck equations (assuming electroneutrality and null current conditions)	Eq. 13
Advection–diffusion equation	Eq. 15

after algebraic manipulation and the introduction of the porosity term (θ), we present the weak formulation, respectively, of Eqs. 3 and 5:

$$\int_{\Omega} \nabla \Phi \cdot \nabla v_{\Phi} dx - \int_{\Omega} \frac{F}{\epsilon} \left(\sum_{i=1}^m z_i c_i + w \right) \cdot v_{\Phi} dx = 0 \quad (6)$$

$$\int_{\Omega} \theta c_i^n v_i dx = \int_{\Omega} \theta c_i^{n-1} v_i - \theta \Delta t D_i (\nabla c_i^n + \frac{z_i F c_i^n}{R T} \nabla \Phi) \cdot \nabla v_i dx \quad (7)$$

where v_{Φ} is the test function associated with the electric potential.

3.1.2 Nernst–Planck equation (concentration-based description of the electric potential)

Assuming that the electroneutrality condition is as follows:

$$\sum_{i=1}^m z_i c_i + w = 0 \quad (8)$$

and the null current condition:

$$F \sum_{i=1}^m z_i J_i = 0 \quad (9)$$

are satisfied, one can describe Φ in terms of concentration of species, as expressed in [92]:

$$\nabla \Phi = - \frac{\sum_{i=1}^m z_i D_i \nabla c_i}{\frac{F}{R T} \sum_{i=1}^m z_i^2 D_i c_i} \quad (10)$$

Combining (7) and (10):

$$\frac{\partial c_i}{\partial t} = - \sum_{k=1}^m D_{ik} \nabla c_k \quad (11)$$

where D_{ik} is the cross-diffusion term and is defined as follows:

$$D_{ik} = D_i \delta_{ik} - \frac{z_i z_k D_i D_k c_i}{\sum_{k=1}^m z_k^2 D_k c_k} \quad (12)$$

where δ is the Kronecker symbol (1 if $i = j$ or 0 otherwise). Finally, we introduce the porosity term and present the weak formulation of Eq. 11:

$$\int_{\Omega} \theta c_i^n v_i dx = \int_{\Omega} (\theta c_i^{n-1} v_i - \theta \Delta t \sum_{k=1}^m D_{ik} \nabla c_k^n) \cdot \nabla v_i dx \quad (13)$$

3.1.3 Advection–diffusion equation

Equation 5 becomes the known advection–diffusion equation, when the electrochemical migration and the source/sink term are neglected, as in [92]:

$$\frac{\partial c_i}{\partial t} = - \nabla \cdot (V c_i - D_i (\nabla c_i)) \quad (14)$$

The weak formulation for a porous medium is then [93]:

$$\int_{\Omega} \theta c_i^n v_i dx = \int_{\Omega} (\theta c_i^{n-1} - \theta \Delta t (V c_i - D_i \nabla c_i^n)) \cdot \nabla v_i dx \quad (15)$$

3.2 Chemical solver

The thermodynamic solver calculates the chemical term $\Gamma(c_1 \dots c_m)$. It describes the equilibrium and kinetic reactions that occur in the fully saturated multicomponent system with an arbitrary number of mineral and gases phases. The chemical equilibrium is calculated with the GEM approach at a given temperature, pressure, and bulk composition, with volume treated as a free parameter. Setup of the chemical system comprises the definition and thermodynamic data for possible solid/liquid/gas phases, temperature, pressure, and the total bulk elemental composition of the system to equilibrate. The newly equilibrated equilibrium state is then used to extract the updated species concentrations.

Reaktoro allows to calculate Lasaga-type kinetics rates [94]. If one needs to describe a special kinetically controlled rate, FEniCS–Reaktoro framework allows one to implement it into the reactive transport loop directly.

In all the benchmarks presented in the following section, the thermodynamic database set up uses the PSI–Nagra database [75].

3.3 Transport and chemical coupling

The chemical and the transport processes are coupled using the sequential non-iterative approach (SNIA) as conceptualized in Fig. 1.

Within each time step, the PDEs for transport of the aqueous components are solved first. The new composition of the chemical system, at each FE, node is passed to Reaktoro for the calculation of thermodynamic equilibria or kinetically controlled chemical reactions. After the chemical equilibration step, the concentrations for each transported species are passed back to each FE node for the next time step.

The transport PDEs are implemented using volumetric concentrations while the chemical solver uses absolute amounts of moles elements. The coupling between transport and chemical solver involves the conversion of

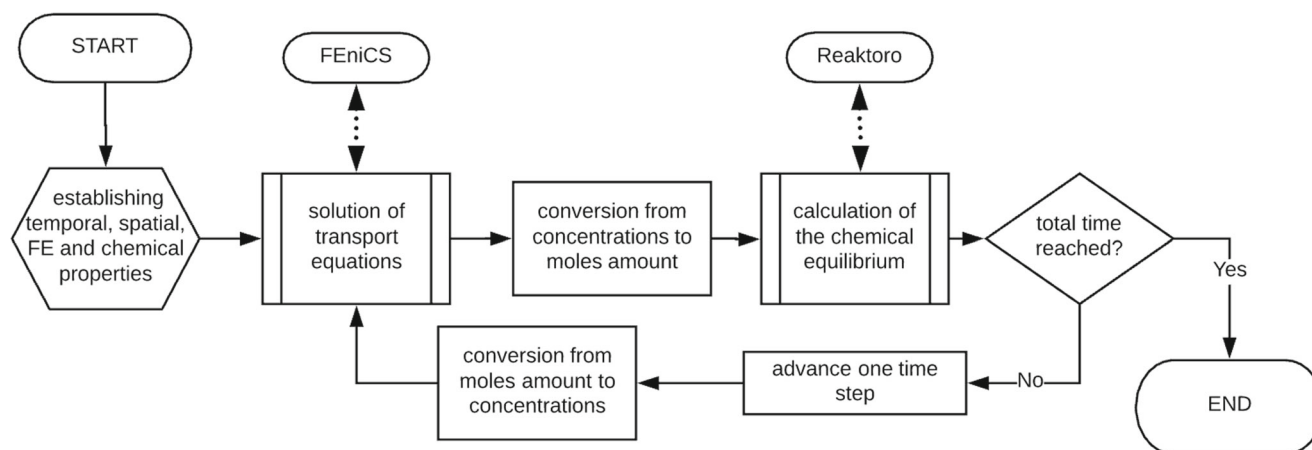


Fig. 1 FEniCS–Reaktoro operator splitting SNIA approach

the volumetric concentrations transported to total moles amounts needed for the chemical solver and backward.

In Appendix A, we present details of the main core class responsible for the transport chemical coupling as well as the UML class and sequence diagram [95].

4 Benchmarking results

In this section, we present four 1D benchmarks that illustrate the code capability to simulate: diffusion dominated flow and advection–diffusion flow, effects of electrochemical migration, mineral precipitation, and dissolution reactions. The results are compared with results from other codes and validated against experimental data. The first benchmark describes the cross-diffusion of ions due to electrochemical coupling. In the second benchmark, simulations of the electrochemical diffusion of ions through an uncharged membrane are compared with experimental data. The third benchmark presents the diffusion of ions through a charged membrane under an external electric field. The fourth benchmark deals with mineral precipitation and dissolution in porous media. Table 3 summarizes the physical processes and the equations used to describe them for each benchmark case.

4.1 Benchmark 1: Coupled electrochemical migration diffusion

This benchmark was developed by [96], inspired by [97]. It describes the multicomponent diffusion of HNO_3 from a $\text{pH} = 4$ solution to a $\text{pH} = 6$ solution, with a constant NaCl background concentration. The left side (at $x = 0$) has a fixed concentration, while on the right side (at $x = 1.0$) an open boundary condition is applied. There are no chemical reactions taking place in this simulation. Table 4 presents the details of the benchmark 1.

Our results show a perfect agreement with the results obtained from Phreeqc and CrunchFlow, as presented in Fig. 2.

4.2 Benchmark 2: coupled electrochemical diffusion through an uncharged membrane

Benchmark 2 represents a simple laboratory experiment similar to benchmark 1. The experiment determines the electrochemical flux of KCl between two reservoirs with an initially identical concentration of KCl . The electrochemical flux of KCl is triggered by the HNO_3 concentration gradient imposed between the two reservoirs. The experimental set up consists of two adjacent reservoirs separated by a cellulose acetate membrane. Cellulose acetate represents an uncharged porous medium. An electroneutral and non-sorbing tracer, HTO (tritiated water), is used to measure the

Table 3 Benchmark descriptions and equations used to describe the transport mechanisms

Benchmark	Description	Eq.
1	Coupled electrochemical migration diffusion	Eq. 13
2	Coupled electrochemical migration diffusion through an uncharged membrane	Eq. 13
3	Coupled electrochemical migration diffusion through a charged membrane with external electric field	Eqs. 6 and 7
4	Mineral precipitation and dissolution based on the advective inflow of MgCl_2 through a calcite column	Eq. 15

Table 4 Chemical composition, diffusion coefficients, and boundary conditions for the electrochemical effect benchmark

Species	Diffusion coefficient [$m^2 \times s^{-1}$]	Boundary condition at $x=0$ [$mol \times m^{-3}$]	Initial condition at $t=10$ [$mol \times m^{-3}$]
H ⁺	9.311×10^{-9}	0.001×10^{-3}	1.0×10^{-4}
NO ₃ ⁻	1.902×10^{-9}	0.001×10^{-3}	1.0×10^{-4}
Na ⁺	1.334×10^{-9}	1.0×10^{-4}	1.0×10^{-4}
Cl ⁻	2.032×10^{-9}	1.0×10^{-4}	1.0×10^{-4}

geometric transport properties (tortuosity) of the membrane. The diffusion of HNO₃ induces an additional flux of KCl to maintain the charge balanced at all times. In the experiment, fluxes of K and Cl against their concentration gradient are observed. This is the so-called “uphill fluxes.” It is when the movement of electrolyte originate from the different mobilities of the involved ions, and initially, such fluxes are controlled by the electrochemical migration term. The differences compared to benchmark 1 are as follows: (1) constantly stirred reservoirs on both sides instead of a constant concentration on the boundary; (2) separation of the solutions by a porous membrane.

To model this experiment properly, a volume adjustment factor was introduced in the weak formulation to account for the different cross-sectional areas and volume sizes of the nodes representing different domains (membrane and tanks). This allows to use a simple 1D FE discretization and no additional discretization of the reservoirs are needed. The membrane porosity is 0.44, the diameter is 23 mm, and the width is 0.13 mm. The domain is discretized into 29 cells, and the volume adjustment factor (α) for membrane cells

is 4.156×10^{-4} while for tanks cells are 1.0. The updated version of Eq. 13 becomes as follows:

$$\int_{\Omega} \alpha \theta c_i^n v_i dx = \int_{\Omega} (\alpha \theta c_i^{n-1} v_i - \alpha \theta \Delta t \sum_{k=1}^m D_{ik} \nabla c_k^n) \cdot \nabla v_i dx \quad (16)$$

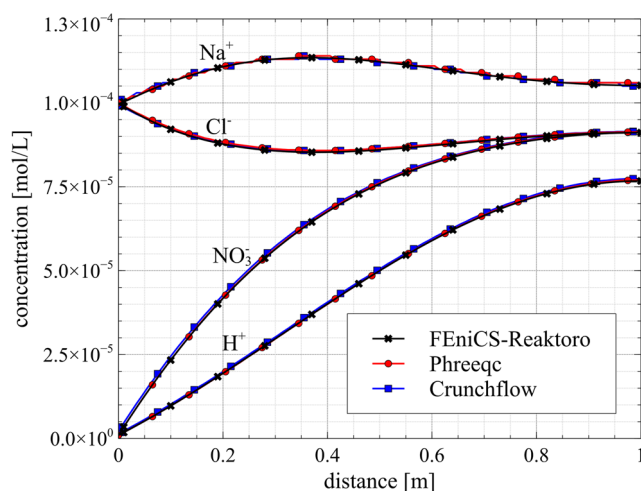
The diffusion coefficients, as well as the chemical composition, are presented in Table 5. The initial condition for the membrane at $t = 0$ is assumed to be the same as the left reservoir.

Figures 3 and 4 present the evolution of the concentration in each reservoir over time. Our results show a very good agreement with Phreeqc, Flotran, and the experimental data. The small differences for temporal evolution of HTO can be explained by the different numerical methods and to the numerical precision among the compared codes.

The electromigration term fully controls the transport phenomena observed in this benchmark [98, 99] and, therefore, the traditional Fickian approach is not sufficient to adequately describe the fluxes observed experimentally, as also discussed in [100].

4.3 Benchmark 3: Coupled electrochemical diffusion through a charged membrane with external electric field

An external electric field is widely used in laboratory experiments [101] to accelerate the transport of charged aqueous species and assist measurements of ion transport properties in a timely manner. In benchmark 3, inspired by [81], the diffusion of ions through a charged membrane is modeled in the presence of an external electric field. The chemical composition and diffusion coefficients are presented in Table 6. The membrane’s porosity is 1.0 and a charge of 50 mol m^{-3} is equally distributed at every membrane node. The magnitude of the applied membrane charge is comparable with the typical surface charge of clay and cement minerals [102].

**Fig. 2** Comparison of the electrochemical benchmark results, inspired in [97] and [96], after 1-h diffusion time**Table 5** Chemical composition, diffusion coefficients, and left and right reservoirs concentrations

Species	Diffusion coefficient [$m^2 \cdot s^{-1}$]	Right reservoir [$mol \times L^{-1}$]	Left reservoir [$mol \times L^{-1}$]
H ⁺	9.0×10^{-9}	0.001×10^{-3}	1.0×10^{-3}
NO ₃ ⁻	1.9×10^{-9}	0.001×10^{-3}	1.0×10^{-3}
K ⁺	$2.0 \cdot 10^{-9}$	$1.0 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$
Cl ⁻	$2.0 \cdot 10^{-9}$	$1.0 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$
HTO	$2.3 \cdot 10^{-9}$	$3.44 \cdot 10^{-10}$	$1.0 \cdot 10^{-15}$

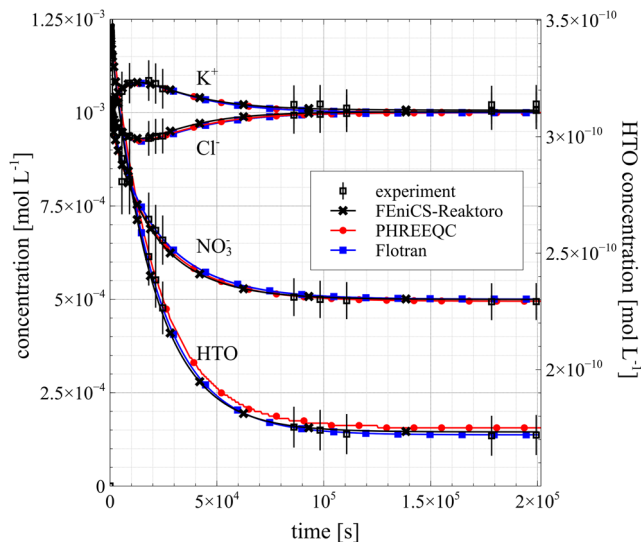


Fig. 3 Left upstream reservoir evolution of reservoir concentration with time for charged species diffusion through a porous uncharged membrane

The external electric gradient of 13 V m^{-1} is oriented from left to right. The dielectric constant of the membrane and the temperature are set to $7.0832 \times 10^{-10} \text{ C V}^{-1} \text{ m}^{-1}$ and 300 K , respectively. The 1 cm thick membrane is discretized into 50 equidistant cells. This benchmark represents the stationary problem obtained from Eq. 7 by setting the change in concentration over time to zero:

$$0 = \int_{\Omega} -\theta D_i (\nabla c_i^n + \frac{z_i F c_i^n}{RT} \nabla \Phi) \cdot \nabla v_i dx \quad (17)$$

Figure 5 presents the comparison between our results and the values calculated by [81] for the stationary case of

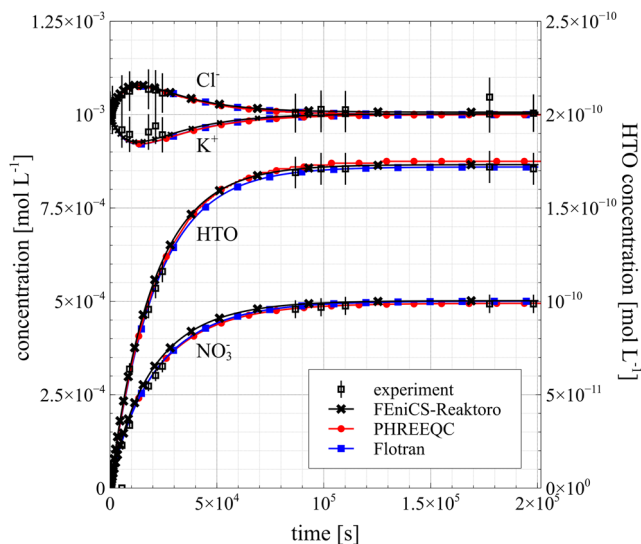


Fig. 4 Right downstream reservoir evolution of reservoir concentration with time for charged species diffusion through a porous uncharged membrane

Table 6 Chemical Composition, diffusion coefficient and boundary conditions

Species	Diffusion coefficient [$\text{m}^2 \cdot \text{s}^{-1}$]	Left boundary condition [$\text{mmol} \cdot \text{L}^{-1}$]	Right boundary condition [$\text{mmol} \cdot \text{L}^{-1}$]
SO_4^{2-}	$3.0 \cdot 10^{-10}$	50.0	41.725
Mg^{2+}	$3.0 \cdot 10^{-10}$	25.0	0.0
Na^+	$3.0 \cdot 10^{-10}$	0.0	8.45
K^+	$5.0 \cdot 10^{-10}$	0.0	25.0

coupled ion diffusion through a charged membrane in the presence of an external electric field.

Ions are forced by the external electric field to move towards the oppositely charged electrode. The concentration of counter-charged ions near the charged surface will be enriched to compensate the electric potential while same-charge ions will be repelled, as observed in clay materials [103]. In a highly compacted medium, as the nanopores in clay interlayers, the counter-ions can be entirely excluded from the pore space between charged mineral surfaces [78]. The obtained results are in perfect agreement with the simulations of [81].

4.4 Benchmark 4: Mineral precipitation and dissolution due to injection of MgCl_2 into a calcite column

This benchmark, originally proposed by [104], is used for the validation of several reactive transport codes [13, 15, 105–107]. The setup consist of a 0.5-m long 1D calcite column which is infiltrated with MgCl_2 through the left side boundary. The initial system porosity is 0.32 and the advective velocity of pore fluid is $9.375 \times 10^{-6} \text{ m s}^{-1}$. A detailed description of the chemical and transport properties

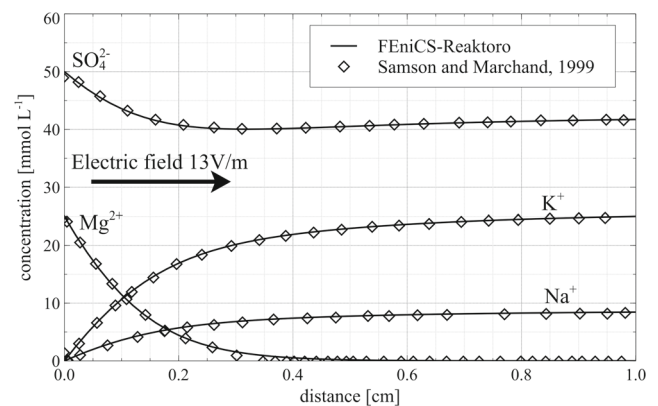


Fig. 5 Results for the stationary diffusion through a charged membrane in the presence of an external electric field

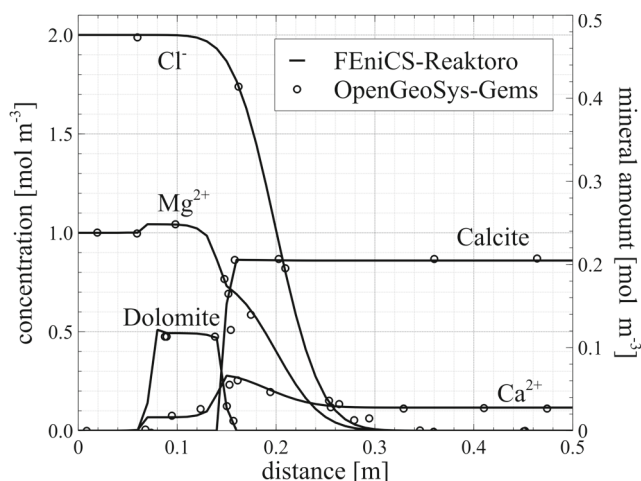


Fig. 6 Comparison of a mineral dissolution-precipitation benchmark where a calcite column is flushed with MgCl_2 causing calcite dissolution and dolomite precipitation

of the system can be found in [106]. The constant inflow of MgCl_2 causes calcite dissolution and temporary dolomite precipitation close to the reaction front as well as an increase of Mg^{2+} in solution. Figure 6 presents the results obtained after 21,000 s of simulation and demonstrates a good agreement between the results of FEniCS-Reaktoro and OpenGeoSys-GEMS.

5 Conclusion

The development of FEniCS-Reaktoro framework was driven by the need for an advanced open-source tool that could provide a flexible modeling environment and, at the same time, allow full control over the numerical methods. FEniCS-Reaktoro was developed using an object-oriented approach and permits the solution of user-defined single or multiphase fluid transport problems in saturated porous media coupled to thermodynamic and kinetic chemical equilibria between a fluid and numerous solid phases. An intuitive math equations alike scripting language allows quick system setup of arbitrary complexity regarding process couplings, domain geometry, and the boundary conditions without need of deep programming skills. All in all, FEniCS-Reaktoro is a goal-oriented and easy to learn framework for reactive transport modeling that combines the computing engine from two state-of-the-art platforms for describing physical and chemical processes.

Several peculiar features of the FEniCS-Reaktoro include the ability to tackle the transport equations at a PDE level allowing adjustments, changes, and implementation of new terms, in a straightforward way. Another aspect of FEniCS-Reaktoro is the combination of electrochemical

migration transport, based on the Nernst–Planck equation with the ability to deal with complex multiphase chemical systems including solid solutions. FEniCS-Reaktoro is compatible with distributed memory HPC architecture offered with PETSc-based parallelization.

The chemical solver, Reaktoro, is probably the most advanced chemical solver available and combines the advantages of existing LMA and GEM thermodynamic solvers. Recently, the standalone numerical kernel of Reaktoro has been significantly improved in numerical performance and convergence speed thanks to machine learning-based algorithms, which help to obtain an initial guess close to the optimal solution.

The open-source FEniCS-Reaktoro framework for reactive transport simulations was described in detail and validated against several conceptually different experimental and numerical benchmarks. Overall, an excellent agreement between the simulation results and the benchmark could be achieved. FEniCS-Reaktoro could reproduce the essential transport relevant processes and chemical reactions consistently demonstrating an enormous potential for further applications.

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Appendix A

The core class of FEniCS-Reaktoro encapsulates two parts: (1) the initialization routine and (2) the reactive transport loop. The initialization routine prepares the underlying structure and defines multiple properties: temporal and spatial; initial conditions, boundary conditions, and constants; single diffusion coefficient or species-dependent diffusion coefficients; the equation system; and the chemical systems. The reactive transport loop is where the transport and chemical equations are solved within a one-time loop until total simulation time is reached.

The finite element mesh and the chemical system are defined during the initialization routine before the reactive transport loop. This process includes the definition of (1) all species and phases involved in the chemical reactions; (2)

governing equations necessary to describe the transport of such species; and (3) initialization of the cells necessary for the domain representation using FE infrastructure provided by FEniCS.

Figures 7 and 8 present the class and sequence diagram, respectively, of the FEniCS–Reaktoro framework using the UML language.

The usage of FEniCS–Reaktoro coupled code to run a reactive transport simulations requires the generation of the following:

- The chemical system file: the chemical system file can be generated using (a) the GEM-Selektor v.3 graphical user interface, (b) the Phreeqc graphical user interface, or (c) Reaktoro’s manual chemical system editor
- The mesh file: the mesh file can be generated using (a) GMSH or (b) FEniCS, depending on the complexity of the geometry
- The 4 files included in the user-defined input package: main input file, transport file, boundary conditions file and diffusion coefficients file

The detailed composition of the 4 files from the user-defined input package is as follows:

- Main input file: contains the temporal and spatial properties, finite element family and degree, type of transport equation used, chemical composition (composed by external chemical definition files), and constants (optional)
- Transport file: contains the description of the physical processes, through the definition of the weak formulation of the PDE
- The boundary conditions file: defines the boundary conditions of the simulation
- Diffusion coefficients file: defines the diffusion coefficient of the species (single or multicomponent diffusion are supported).

We point to the framework’s repository located at bitbucket.org/lhdamiani/fenics-reaktoro, where one can find instructions, licensing information, documentation, and demos.

Fig. 7 The UML class diagram description of FEniCS–Reaktoro framework

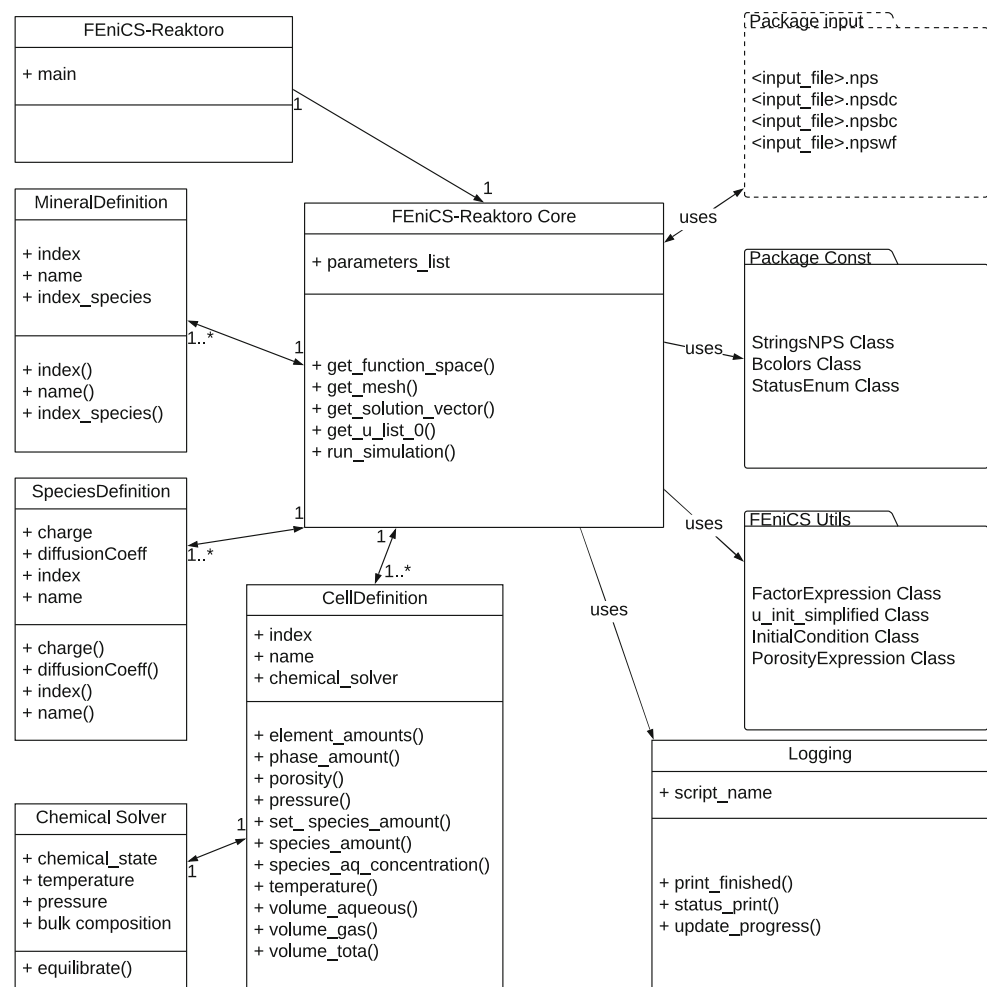
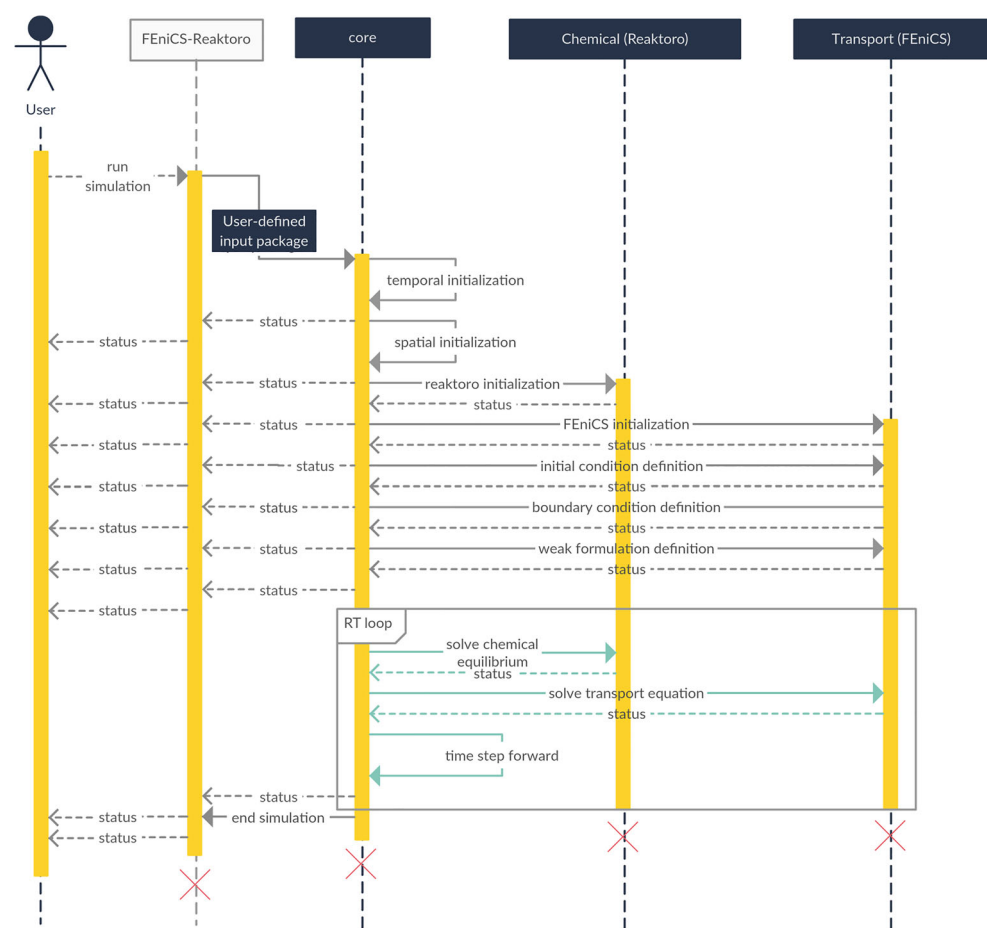


Fig. 8 The FEniCS-Reaktoro framework UML sequence diagram depicting the interaction between the components



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